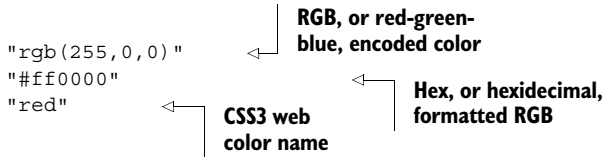
(continued)

It's worth studying the properties of color—hue, value, intensity, and temperature—to ensure the most harmonious color relationships in a visualization. Leonardo da Vinci organized colors into psychological primaries, the colors the eye sees unmixed, but the modern exploration of color theory, as with many other phenomena in physics, can be attributed to Sir Isaac Newton. Newton observed the separation of sunlight into bands of color via a prism in 1666 and called it a color spectrum. Newton also devised a color circle of seven hues, a precursor to the many future visualizations that would organize colors and their relationships. About a century later, J. C. Le Blon identified the primary colors as red, yellow, and blue, and their mixes as the secondaries. The work of other more modern color theoreticians like Josef Albers, who emphasized the effects of color juxtaposition, influences the standards for presentation in print and on the web.

Color is typically represented on the web in red, green, and blue, or RGB, using one of three formats: hex, RGB, or CSS color name. The first two represent the same information, the level of red, green, and blue in the color, but do so with either hexadecimal or comma-delimited decimal notation. CSS color names use vernacular names for its 140 colors (you can read all about them at http://en.wikipedia.org/wiki/Web_colors#X11_color_names). Red, for instance, can be represented as



D3 has a few helper functions for working with colors. The first is `d3.rgb()`, which allows us to create a more feature-rich color object suitable for data visualization. To use `d3.rgb()`, we need to give it the red, green, and blue values of our color:

```
teamColor = d3.rgb("red");
teamColor = d3.rgb("#ff0000");
teamColor = d3.rgb("rgb(255,0,0)");
teamColor = d3.rgb(255,0,0);
```

These color objects have two useful methods, `.darker()` and `.brighter()`. They do exactly what you'd expect: return a color that's darker or brighter than the color you started with. In our case, we can replace the gray and red that we've been using to highlight similar teams with darker and brighter versions of pink, the color we started with: